1. A 5 ml. sample of waste water was refluxed with 30 ml. of potassium dichromate solution and after refluxing, the excess unreacted dichromate required 23 ml. of 0.1 M F.A.S. solution. A blank of 5 ml. of distilled water on refluxing with 30 ml. of dichromate solution required 36 ml. of 0.1 M F.A.S. solution. Calculate the C.O.D. value of the waste water.

$V_{blank} = 36\,ml$ $N_{FAS} = 0.1M \times 0.1 = 0.1$

$V_{back} = 23\,ml$ $V = 5\,ml$

the formula for $COD = \dfrac{(V_2 - V_1)_{FAS}\, N_{FAS} \times 8 \times 1000}{V}$ ppm

$= \dfrac{(36 - 23)\,0.1 \times 8 \times 1000}{5} = 2080$ PPM

$$\boxed{E\ factor = \frac{\text{Amount of waste}}{\text{Amount of Product}}}$$

A chemical synthesis produces 3.86 g of product and generates 10.89 g of waste (including solvents and byproducts). Calculate the E-factor.

$E\ factor = \dfrac{\text{Amount of waste}}{\text{Amount of Product}} = \dfrac{10.89}{3.86} = 2.82 = E$ factor.

13.A reaction uses 3.2 g of resin, 0.372g of catalyst (TO BE EXCLUDED), and produces 3.5 g of intermediate product. Calculate the E-factor.

$E\ factor = \dfrac{\text{Amount of waste}}{\text{Amount of product}} = \dfrac{3.2 - 3.5}{3.5} = 0$ gm

14) 100 ml wastewater containing 920 ppm dissolved oxygen was diluted with 100 ml distilled water and kept in a bottle at 20°C for 5 days. The oxygen content of the resulting water was then found to be 260 ppm. Calculate the BOD of the sample.

formula for BOD = $(DO)_1 - (DO)_2$

$DO_1 \rightarrow$ Dissolved oxygen in water before incubation
$DO_2 \rightarrow$ Dissolved oxygen in water after incubation

$v = 100ml \rightarrow 0.1l$. $(DO)_1 \rightarrow 420$ ppm $(DO)_2 \rightarrow 260$ppm

$\Rightarrow \frac{(DO_1) - (DO)_2}{0.1} \Rightarrow \frac{420 - 260}{0.1}$

↳ Doubt in this or to is this the correct method.

15) 5 ml of raw sewage was diluted by specially prepared water in a 300 ml capacity BOD bottle. The DO concentration of the diluted sample at the beginning of the test was 9 mg/l and 6 mg/l after 5-day incubation at 20°C. Find the BOD of raw sewage.

formula $\Rightarrow$ BOD = $(DO)_1 - (DO_2)$. $(DO)_1 = 9$ & $(DO_2) = 6$mg/l.

The BOD = 9-6 = 3mg/l of the raw sewage.

firstly you'd to calculate the DF or the dilution factor

i.e Dilution factor (DF) = $\frac{\text{volume of bottle}}{\text{volume of sample}}$ =

$\therefore (DF) = \frac{300}{5} = 60$ mg/l.

now BOD $= ((DO)_1 - (DO)_2) \times DF$

$= (9 - 6) \times 60$
$= 180$ mg/l

$\therefore$ The raw BOD of the sewage is 180mg/l.

16) In order to conduct a 5-day BOD test, the sample wastewater was diluted with specially prepared dilution water, with a dilution factor of 150. The contents of dissolved oxygen in the beginning and end of the test were found to be 11 ppm and 7 ppm respectively. Compute the 5-day BOD and comment on the nature of the wastewater sample?

(DF) Dilution factor $= 150$

$(DO)_1 = 11$ ppm , $(DO)_2 = 7$ ppm

$$BOD = (11 - 7) \times 150 = 600$$

BOD strength is considered along the following term.

factror

weak	< 100
Medium	> 100 < 500
strong	> 500

So yes it's very strong.

16) In order to conduct a 5-day BOD test, the sample wastewater was diluted with specially prepared dilution water, with a dilution factor of 150. The contents of dissolved oxygen in the beginning and end of the test were found to be 11 ppm and 7 ppm respectively. Compute the 5-day BOD and comment on the nature of the wastewater sample?

Understanding nitrate Sources :-

Nature → Antropogenic (human caused)

① Nitrate Contamination

Primary Sources

- Agricultural practices (overuse of urea, pesticides)
- Improper Sanitation System (leaking of Septic tank, poor sewage)
- Industrial Sources (Storage tanks, landfills)
- Environmental degradation (deforestation → loss of natural filter)

② Health Impacts

Infants (very high Risk)

- Blue baby Syndrome
- Reduced O_2 transport → bluish Skin, Respiratory failure.

③ Groundwater availability & Extraction trends

Extraction Rate (ExoR)

- Steady at 60.4% Since 2009

Zonal Improvement (ZM)

- 73% of blocks now "Safe" (up from 67.2% in 2022)

Over exploitation (Exploits)

- uranium concentration higher in deep water Zones.

④ Institutional Framework

Establishment

- founded in 1970 under Ministry of water resources (now Jal Shakti)

Role

Apex body for groundwater management, exploitation, monitoring, regulation.

Regulatory Arm

- CGWA (under EPA 1986)
- Issues Extraction permits

Key Strategies

- Artificial recharge schemes.
- Rainwater harvesting promotion
- Publishing hydrological reports & atlases.

① Critical term for Revision

Leaching

- chemicals dissolve in rainwater → seep into underground.

Aquifier

- underground water bearing rock | material layer.

Dark Zone.

- Extraction > Annual recharge.

Nitrates permissible limits 45mg/L (BIS/WHO)

② Fluoride – Double edged Sword

- Safe limit ≤ 1.5mg/L
- Geographical Spread (RTAH)

Health Impacts

- Dental fluorosis – staining/pitting of teeth

- Skeletal fluorosis – Joint pain stiffness, deformities.

Sources
- weathering of the fluride rich rocks (fluoride, Apathite)
- overextraction → Higher Concentration.

③ Arsenic – "Silent Killer"

Safe Limit ≤ 0.01mg/L

Geographical Spread: Ganga Brahmaputra plains (WB, UP, Bih, Ass)

Health Impacts
- Arsenocoisis → Skin lesions, "raindrop" pigmentation
- Carcenogenic → Skin, lung, bladder Cancers.

Sources
- Himalayan Sediments
- released via chemical reaction, worsned by irrigation pumping

④ Aluminium

Safe Limit 0.03 – 0.2mg/L

Geographical Spread (KOWB) acidic laterite soil.

Health impacts
- Neurotoxicity → Alzimer's, parkinson
- Bone fragility → Interferes with Calcium absorption.

Sources
- Geogenic (bauxite) laterite rocks)
- Acid rain, industrial waste (Smelting, Paper Mills)

⑤ Mitigation Strategies

① Alternative Sources (Surface water supply)
② Fluride removal (Nalgonda technique) → alum + lime
③ Arsenic removal | Community Aru's) → Precipiate As chemically at hand pumps.
④ Dilution (rainwater harvesting)

land degradation & desertification :-

1. Core Concept.

Definition :- Decline) loss of land productivity due to human activity + climate variation.

Scope

- physical → soil structure deteriotion
- chemical → nutrient loss salinization
- Biological → microbial health decline.

2. Major Consequences (FHBS)

- Food Security → reduced soil moisture & fertility → lower crop yeilds
- Hydrological Crisis → reduced ground water recharge (eg bengaluru Crisis)
- Biodiversity loss → habitat destruction (eg extinction of pink headed duck
- Socio economic Shift → rural migration (eg odisha) → urban centres

③ Traditional Ecological Knowledge.

Zabo System (Nagaland)

mechanism ◦ Rain water Collected → Ponds → Cattle Sheds (nutrients) → terraced Rice fields

Benifit { ◦ Benifits → Prevents erosion, maintain fertility, ensure water avilability.